Abubakar Khawaja

AI Engineer | LLM Applications | RAG | Agents | Python | TypeScript

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PROFESSIONAL SUMMARY

AI Engineer with 5+ years of software engineering experience, now building LLM applications on top of production Python and TypeScript systems. I have shipped streaming chat interfaces with the Vercel AI SDK, agentic tool calling where the model picks between a pgvector knowledge base and live web search, and RAG retrieval over embeddings I generate and index myself. I built an MCP server on Anthropic's Model Context Protocol SDK that exposes 12 zod-validated tools, enough for an agent to run a full workflow unattended. I am comfortable owning an AI feature from retrieval quality and context-window management through to the interface it renders in.

EXPERIENCE

Senior Software Engineer | Codeupnow Sept 2025 – Present

- Worked on a microservices system in Python and Django with a React frontend, and cut downtime by 30%.
- Built real-time data pipelines and background job processing for work that runs asynchronously.
- Reworked Python REST API code and cut response times by 40%, which also brought cloud costs down.
- Deployed and tuned services on AWS ECS/EKS and Lambda to support scaling, build performance, and deployment reliability.
- Optimized frontend performance with lazy loading, code splitting, and image optimization.

Senior Full-Stack Engineer | Arbisoft July 2021 – Apr 2026

- Used AI coding tools such as Cursor, Claude Code, Codex, GitHub Copilot, and ChatGPT for both autonomous multi-step tasks and hands-on pair-programming, applying them to PR review, architecture discussions, and custom internal tooling, while comparing model outputs using structured prompts.
- Designed secure REST APIs with Django REST Framework, implementing JWT authentication, authorization, and third-party integrations.
- Elevated automated test coverage from 10% to over 60% (Jest, Cypress, Vitest) and integrated quality gates (ESLint, Husky), reducing production bugs by 25%.
- Reworked GitLab CI/CD pipelines with parallel jobs and layer caching, reducing deployment times by 40%.
- Owned Sentry monitoring to cut production error volume: error triage with session replay, release and performance tracking, Slack and GitHub integration, and PII-safe payload encryption that keeps sensitive data out of captured events.

SELECTED PROJECTS

Landit, Career Platform · *React + Next.js + TypeScript*

- Built a streaming retrieval chat feature backed by pgvector embeddings, rendering model output token-by-token over semantically retrieved content.
- Built a custom visualization layer for Looker in React after Looker's support team said it could not be done, hosting the built bundles and registering them through Looker config so they appeared in the standard visualization dropdown alongside the built-in charts.
- Executed zero-downtime frontend framework migrations (Material UI v4 to v5, React Router v4 to v5).

Telehealth Platform · *Next.js + Django + PostgreSQL + Kubernetes · Codeupnow*

- Architected and maintained a HIPAA-compliant telehealth platform on Django, with PostgreSQL on AWS RDS and Kubernetes for container orchestration.
- Built the frontend on the Next.js App Router with React Server Components, keeping data fetching on the server.
- Optimized patient-provider matching workflows with SQL query tuning and Redis caching, and worked with data engineering on ETL pipelines for real-time data processing.

Retrieval Chat, Agentic AI Assistant · *Vercel AI SDK + OpenAI + Supabase pgvector · Personal Project*

- Implemented agentic tool calling where the model selects between a custom pgvector knowledge-base retriever and live web search, surfacing the chosen source alongside the answer.
- Built a streaming chat interface with the Vercel AI SDK, rendering model output token-by-token with markdown formatting and sanitised HTML.
- Managed context-window pressure by summarising older conversation turns while preserving recent history.
- Ran object detection client-side with Transformers.js, drawing confidence-filtered bounding boxes in the browser.

Resume Toolkit, MCP Server & CV Tailoring Engine · *Node.js + Model Context Protocol SDK · Open Source*

- Built a stdio MCP server on Anthropic's Model Context Protocol SDK, exposing 12 tools with zod-validated inputs that let an AI agent drive resume tailoring end to end.
- Enforced three tailoring modes at the tool boundary, rejecting any generated text that does not trace back to an approved entry.
- Added gap reporting that surfaces job requirements with no supporting evidence, aggregated across applications to show what is repeatedly missing.

CORE SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, SQL, HTML5, CSS3/SCSS
AI/ML: LLM Application Architecture, AI Agents, Tool/Function Calling, Structured Outputs (Zod, Pydantic), Model Context Protocol (MCP), RAG, Embeddings & Vector Databases (pgvector), Prompt & Context Engineering, OpenAI / OpenRouter / Ollama (local models), Vercel AI SDK
AI Development: ChatGPT, GitHub Copilot, Cursor, Claude Code, Codex, Prompt Engineering, AI-Assisted Development
Backend: Python (Django, Django REST Framework, FastAPI, Flask), Node.js (NestJS, Express), REST API design, GraphQL, webhooks, JWT & OAuth2 authentication, Celery & RabbitMQ (async jobs), WebSockets, Scrapy
Frontend: React.js, Next.js (App Router, Pages Router), Redux Toolkit, TanStack Query, React Hook Form, Tailwind CSS, Material UI, SCSS
Databases: PostgreSQL, pgvector, MySQL, MongoDB, Redis, Supabase, Drizzle ORM & Django ORM, Database Indexing, Query Optimization, Caching
Cloud & DevOps: Vercel, Supabase, AWS (EC2, S3, RDS, ECS/EKS, Lambda), Docker, GitHub Actions, GitLab CI/CD, Sentry, Nginx, Linux, Git
Testing & Quality: Jest, Cypress, Vitest, Storybook, ESLint, Prettier, Husky (pre-commit hooks)
Integrations: Stripe (payments), SendGrid & Resend (email), Firebase Auth, Google Maps, PostHog & Google Analytics, Google Looker (BI), Grafana, JWT/OAuth2
Foundations: Data Structures & Algorithms, System Design, Performance Optimization, Agile/Scrum, Jira & Confluence, Code Review, Mentoring

EDUCATION

B.S. Software Engineering Punjab University College of Information & Technology (PUCIT) · GPA: 3.08

CERTIFICATIONS

- The AI Engineer Path & Prompt Engineering for Web Developers — Scrimba (July 2026)